

Ruoyun Li

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EDUCATION

Yale University M.S. in Public Health in Health Informatics	08/2025-05/2027
University of Virginia (UVA) B.S. in Statistics & Psychology, Highest Distinction	08/2021-05/2025 GPA: 3.95/4.00

RESEARCH INTERESTS

High-dimensional data analysis, Electronic Health Records, Brain Connectomics and Neuroimaging, Population Health

THESIS & PUBLICATIONS

Li, R., Sequeira, S. & Zhang, A. "Latent Dimensions of Social Determinants of Health and Their Associations with Youth Psychopathology Over Time." Manuscript in preparation.

Liu, P., **Li, R.**, Zhao, Y., et al. "A Cost-aware Framework for Personalized Feature Acquisition in Multi-Disease Prediction." Manuscript in preparation.

Wang, T., Lin, J., Zhang, X., Xiao, W., Li, Y., Chen, Y., Peng, J., Xing, H., **Li, R.**, et al. "HealthResearchBench: Reusable Test Beds for Reproduction and Independent Improvement in Healthcare and Biomedical Research." Submitted to AAAI 2027, under review.

Li, R., "New Places, New Feelings: How Primal World Beliefs and Mobility Behaviors Predict Momentary Subjective Experiences." Distinguished Majors Thesis in Psychology, University of Virginia, Advisor: Prof. Adrienne Wood & Dr. Sareena Chadha

CONFERENCE PRESENTATIONS

Li, R., Sequeira, S. & Zhang, A. (2026, June). High-Dimensional Mediation of Social Determinants of Health and Youth Psychiatric Symptoms via Functional Connectivity. Poster presented at the Organization for Human Brain Mapping (OHBM) Annual Meeting, Bordeaux, France

Li, R., Wood, A., & Chadha, S. (2025, April). New Places, New Feelings: How Primal World Beliefs and Mobility Behaviors Predict Momentary Subjective Experience. Poster presented at the L. Starling Reid Undergraduate Psychology Research Conference, Charlottesville, VA

RESEARCH EXPERIENCE

Post-Training and Data Filtering for a Medical Foundation Model 07/2026 – Present
Supervisor: Prof. Hua Xu, Department of Biomedical Informatics & Data Science, Yale School of Medicine New Haven, CT

- Developed data filtering and post-training strategies for a medical foundation model trained on longitudinal claims data, focusing on high-quality cohort construction and instruction fine-tuning
- Designed EHR-specific information-density and data-quality filters to prioritize clinically informative patient trajectories

HealthResearchBench: Reusable Test Beds for Healthcare and Biomedical Research 05/2026 – Present
Collaborative Group Research Project

- Contributed to the experimental schema and evaluation protocol for agent-based reproduction of healthcare and biomedical research
- Guided AI agents in reproducing TransUNet medical image segmentation and the MIMIC-III multitask benchmark, auditing experimental setup, data processing, model execution, and reproduced results for scientific validity
- Analyzed agent performance and failure modes across reproduction experiments; co-authored the resulting manuscript submitted to AAAI 2027

Causal Mediation of Physical Activity and Disease Risk through Organ Imaging in UK Biobank 07/2025 – Present
Supervisor: Prof. Yize Zhao, Department of Biostatistics, Yale School of Public Health New Haven, CT

- Integrated accelerometer, multi-organ imaging, longitudinal disease, and GWAS data from UK Biobank to study how physical activity influences disease risk through organ-level pathways
- Built curated multimorbidity outcomes using ICD-to-PheCode mapping and chronic disease filtering
- Constructed disease hyperedge outcomes using longitudinal PheCode co-occurrence networks to capture higher-order comorbidity structure
- Applied interventional mediation and Mendelian randomization to estimate organ-specific causal effects using survival models and bootstrap-based inference

Cost-Aware Framework for Personalized Feature Acquisition in Multi-Disease Prediction 07/2025 – Present
Supervisor: Prof. Yize Zhao, Department of Biostatistics, Yale School of Public Health New Haven, CT

- Developed a UK Biobank data preprocessing pipeline integrating 1,773 predictors across lifestyle, clinical measurements, environmental exposures, genetics, and imaging, with PCA-based dimensionality reduction and time-aligned disease outcome construction for downstream multimodal modeling
- Mapped UK Biobank features to CPT/HCPCS codes and real-world reimbursement data to estimate acquisition costs

High-Dimensional Mediation of Social Determinants, Brain Connectivity & Youth Mental Health 01/2025 – 08/2026

Supervisor: Prof. Aiying Zhang, NeuroBioInformatics Lab, School of Data Science, UVA

Charlottesville, VA

- Conducted high-dimensional multivariate mediation analyses in the ABCD cohort to quantify how whole-brain resting-state functional connectivity (rsFC) mediates associations between social determinants of health and youth psychiatric symptom severity
- Reduced multi-domain SDOH variables into interpretable latent components using sparse PCA in R, enabling stable modeling under high-dimensional settings
- Implemented the Directions of Mediation (DM) framework in MATLAB to model whole-brain connectivity as a high-dimensional mediator, with bootstrap-based inference to estimate indirect effects and statistical significance
- Applied longitudinal linear mixed-effects models to examine associations between SDOH latent dimensions and changes in OCD, anxiety, and depressive symptoms over time

Longitudinal Modeling of Mobile Sensing, EMA, and Psychological Well-Being

08/2023 – 05/2025

Supervisor: Prof. Adrienne Wood, Emotion and Behavior Lab, Department of Psychology, UVA

Charlottesville, VA

- Completed a Distinguished Majors Program thesis investigating how primal world beliefs and social exploratory behaviors influence psychological well-being over time, using mobile sensing data.
- Processed 1.2 GB of mobile-sensing data through data cleaning, preprocessing, and visualization using Pandas, NumPy, and Matplotlib in Python.
- Analyzed longitudinal data using linear mixed-effects models implemented in R with the lme4 package, identifying temporal relationships among world beliefs, exploratory behavior, and psychological well-being.

PROFESSIONAL SKILLS

- **Biomedical & Health Informatics:** Electronic health records, Medical imaging data, Wearable and behavioral data, Multimodal data analysis
- **Programming Languages:** Python, R, SQL, MATLAB
- **Language:** English (Proficient), Chinese (Native)